

Ex.no:8	MINI PROJECT – Air Quality Analysis

Aim:

To clean, transform, store, and analyze hotel booking data using Python, MySQL, Power BI and Tableau to derive meaningful insights.

Software and Tools Required

- Python
- VS Code
- Pandas
- SQL Server
- SQL Server Management Studio (SSMS)
- Tableau Desktop
- TabPy

3. Process / Procedure

Create a project folder:

AirQualityAnalysis

```
|
|— .venv
|— powerbi.py
|— AirQuality_20k_Insert.sql
|— india_air_quality_consolidated.csv.xlsx
```

Create and activate the Python virtual environment.

Python and Pandas are used for data cleaning and preprocessing.

Open SQL Server Management Studio (SSMS) and connect using:

Server: localhost

Authentication: Windows Authentication

Create the database:

CREATE DATABASE AirQualityProject;

GO

Create the AirQuality table:

USE AirQualityProject;

GO

CREATE TABLE AirQuality

(

city VARCHAR(50),

location VARCHAR(100),

date DATE,

pm25 FLOAT,

pm10 FLOAT,

o3 FLOAT,

no2 FLOAT,

so2 FLOAT,

co FLOAT

);

GO

The table contains:

- City
- Location
- Date
- PM2.5
- PM10
- O3
- NO2
- SO2
- CO

Clean and Prepare Dataset Using Python

Python and Pandas are used to:

- Remove duplicate records
- Handle missing values
- Convert dates into the correct format
- Convert pollutant values into numeric format
- Select exactly **20,000 records**

The original dataset contains approximately **96,755 records**.

The cleaned dataset is prepared for storage in SQL Server.

Insert Data into SQL Server

The Python program connects to SQL Server and inserts the cleaned 20,000 air-quality records into the database.

The data flow is:

Python / Pandas



Air Quality Dataset



Data Cleaning



SQL Server



AirQualityProject



AirQuality Table



20,000 Records

Verify the Dataset

Run:

USE AirQualityProject;

GO

```
SELECT COUNT(*) AS total_records
```

```
FROM AirQuality;
```

Result:

```
total_records
```

```
-----
```

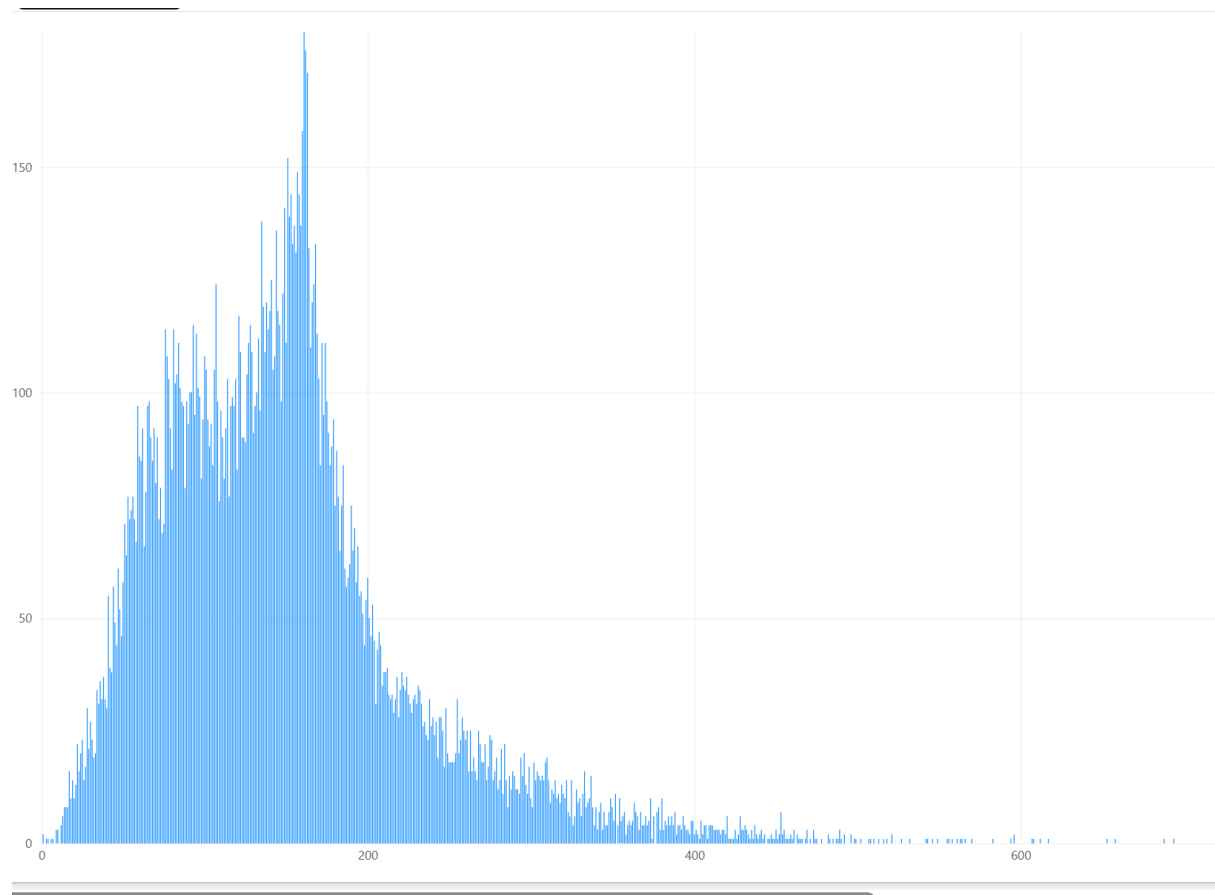
```
20000
```

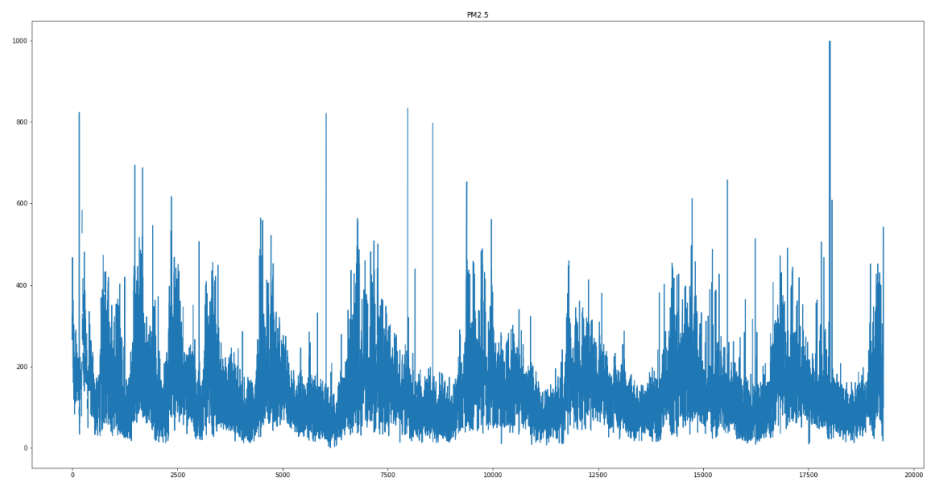
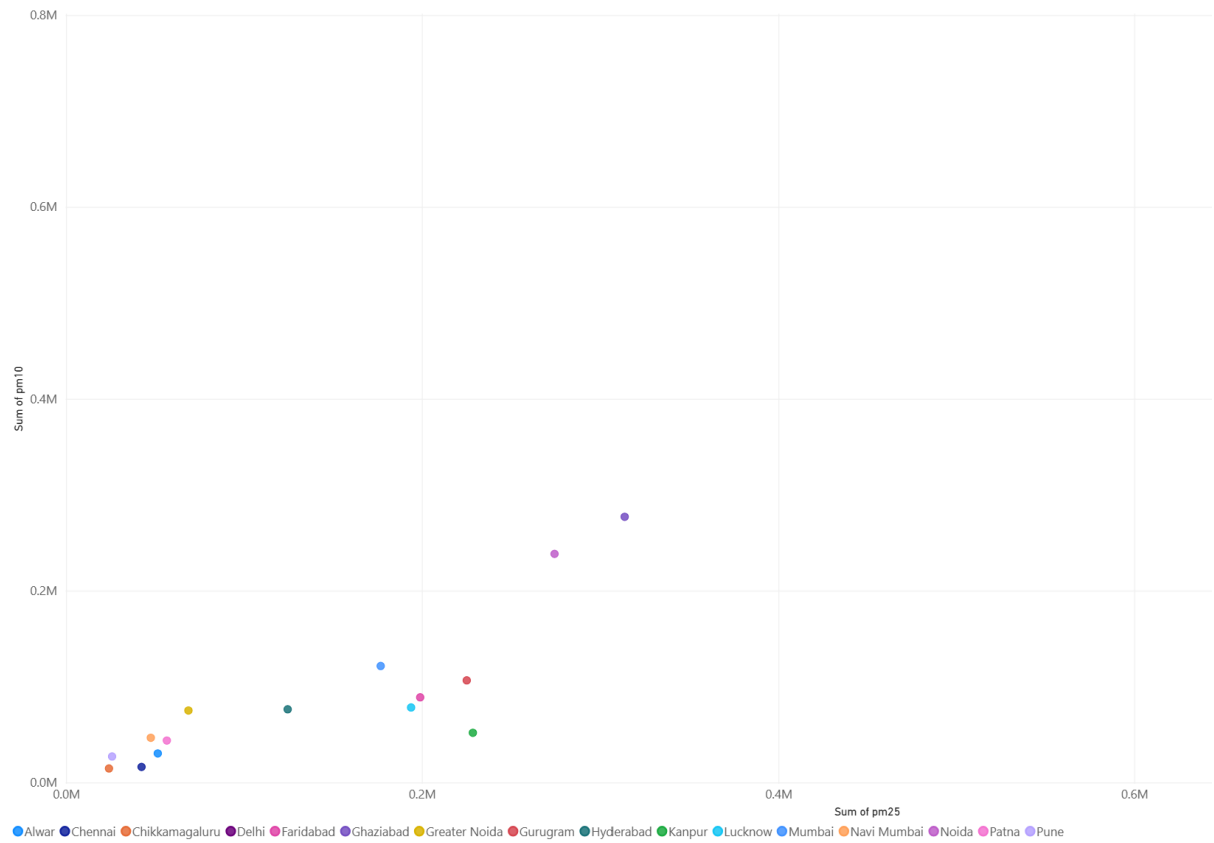
This confirms that all **20,000 air-quality records** have been successfully stored in SQL Server.

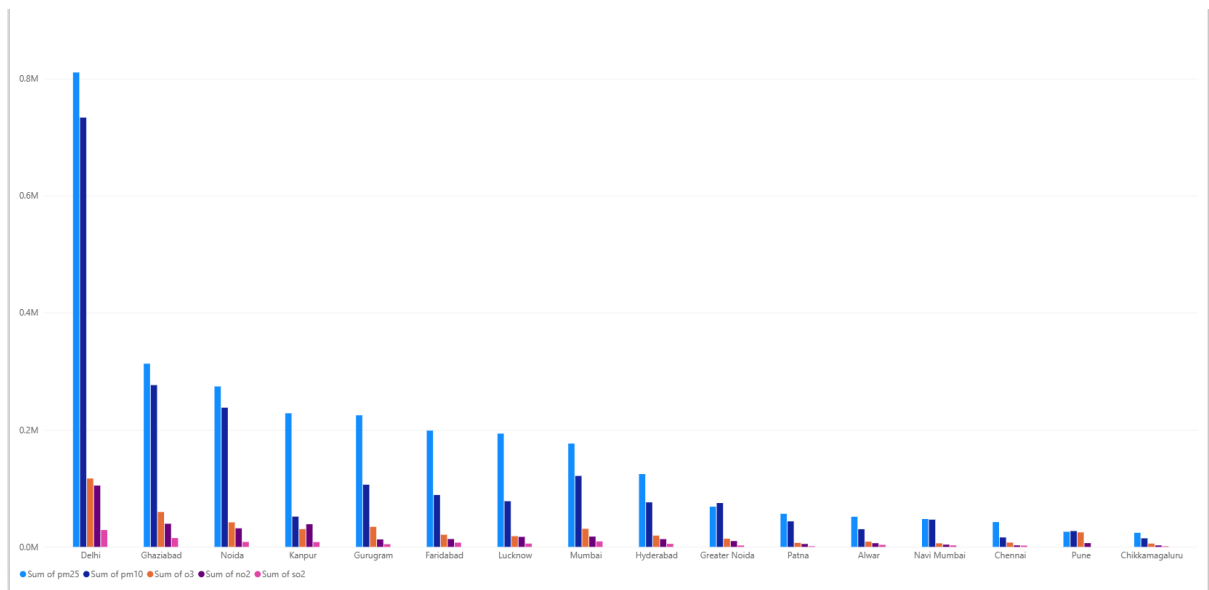
Total Records							
1	20000						
Total Records		Missing_PM25	Missing_PM10	Missing_O3	Missing_NO2	Missing_SO2	Missing_CO
1	20000	702	3213	1317	1334	3406	1658
Average_PM25		Average_PM10	Average_O3	Average_NO2	Average_SO2	Average_CO	
1	148.447662970256	120.648954548162	23.9026922871059	17.4749276759884	6.3231891044956	11.0810707665467	
city		Record_Count	Average_PM25				
1	Delhi	4523	183.78445149592				
2	Gurugram	1400	165.372980910426				
3	Faridabad	1258	164.929577464789				
4	Ghaziab...	1954	162.314182194617				
5	Greater ...	447	157.12100456621				
6	Kanpur	1615	154.828029790115				
7	Lucknow	1312	154.548207171315				
8	Noida	1823	154.521665728756				
city		Record_Count	Average_PM10				
1	Greater Noida	447	169.386004514673				
2	Delhi	4523	166.079918496717				

Visualizations

POWER BI:

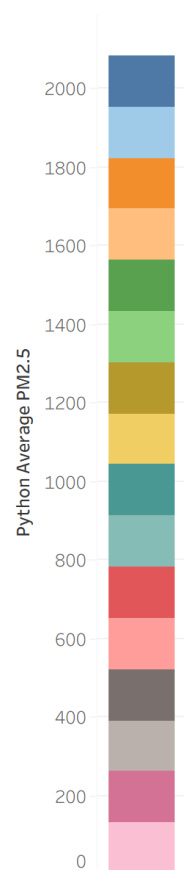




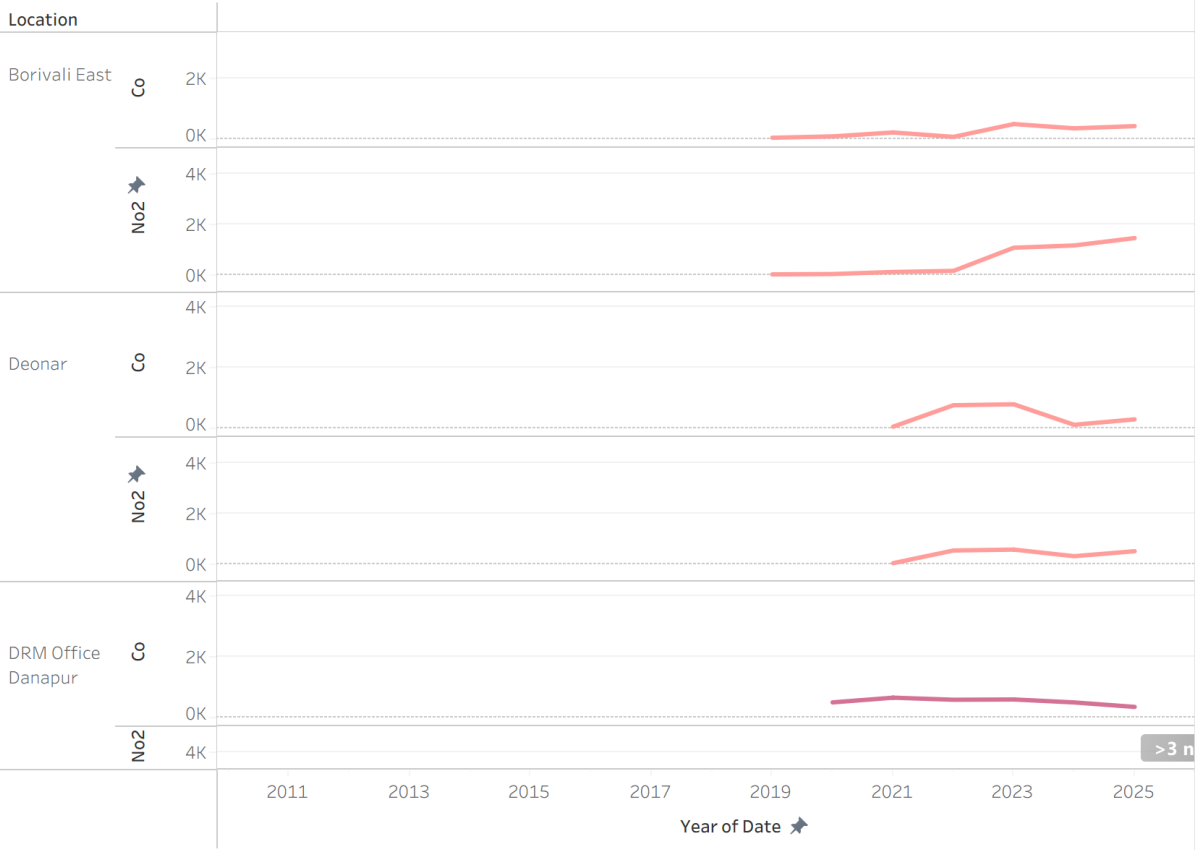


TABLEAU

Sheet 1



Sheet 2



City	Location	
Alwar	Moti Doongri	
Chennai	Velachery Residential Area	
Chikkamagaluru	Kalyana Nagara	
Delhi	Anand Vihar	

Result:

The Visualizations from Power BI is done successfully.

Ex.no:9	MINI PROJECT – Mobile Phone Sales Analysis

Aim:

To clean, transform, store, and analyze hotel booking data using Python, MySQL, Power BI and Tableau to derive meaningful insights.

Software and Tools Required

- Python
- VS Code
- Pandas
- SQL Server
- SQL Server Management Studio (SSMS)
- Power BI Desktop

3. Process / Procedure

Step 1 — Project Setup

Create a project folder:

MobilePhoneSalesAnalysis

```
|
|— .venv
|— mobile_sales.py
|— MobileSales_Project_SQL_20k.sql
|— cleaned_mobile_phone_sales_data.csv.xlsx
```

Create and activate the Python virtual environment.

Python and Pandas are used for data cleaning and preprocessing.

Create Database

Open SQL Server Management Studio (SSMS) and connect using:

Server: localhost

Authentication: Windows Authentication

Create the database:

```
CREATE DATABASE MobileSalesProject;  
GO
```

Create Table

Create the MobileSales table:

```
USE MobileSalesProject;  
GO
```

```
CREATE TABLE MobileSales  
(  
    Product VARCHAR(200),  
    Brand VARCHAR(100),  
    [Product Code] VARCHAR(100),  
    [Product Specification] VARCHAR(500),  
    Price FLOAT,  
    [Inward Date] DATE,  
    [Dispatch Date] DATE,  
    [Quantity Sold] INT,  
    [Customer Name] VARCHAR(200),  
    [Customer Location] VARCHAR(200),  
    Region VARCHAR(100),  
    [Processor Specification] VARCHAR(200),  
    RAM VARCHAR(50),  
    ROM VARCHAR(50)  
);  
GO
```

The table contains:

- Product
- Brand
- Product Code
- Product Specification
- Price
- Inward Date
- Dispatch Date
- Quantity Sold

- Customer Name
- Customer Location
- Region
- Processor Specification
- RAM
- ROM

Clean and Prepare Dataset Using Python

Python and Pandas are used to:

- Remove duplicate records
- Handle missing values
- Convert dates into the correct format
- Convert price and quantity into numeric values
- Select exactly **20,000 records**

The original dataset contains **24,983 records**.

The cleaned dataset is prepared for storage in SQL Server.

Insert Data into SQL Server

The Python program connects to SQL Server and inserts the cleaned **20,000 mobile phone sales records** into the database.

The data flow is:

Python / Pandas



Mobile Sales Dataset



Data Cleaning



SQL Server



MobileSalesProject



MobileSales Table



20,000 Records

Verify the Dataset

Run:

USE MobileSalesProject;

GO

SELECT COUNT(*) AS total_records

FROM MobileSales;

Result:

total_records

20000

This confirms that all **20,000 mobile phone sales records** have been successfully stored in SQL Server.

Total_Records																			
1	20000																		
Product	Brand	Product Code	Product Specification	Price	Inward Date	Dispatch Date	Quantity Sold	Customer Name	Customer Location	Region	Processor Specification	RAM							
1	Mobile Phone	Motorola	88EB4558	Site candidate activity company there bit inside sold...	78570	2023-08-02	2023-08-03	6	William Hess	South Kelsey	Central	Snapdragon 7 Gen	12GB						
2	Mobile Phone	Samsung	9F975B08	Energy special low seven place audience.	159826	2025-03-19	2025-03-20	5	Leah Copeland	South Todd	Central	MediaTek Dimensity	8GB						
3	Mobile Phone	Dell	14932CAE	Could before almost wind couple evening perform...	11670	2023-10-10	2023-10-16	6	Nicole Gonzalez	South Miguel	North	Apple A-Series	16GB						
4	Mobile Phone	Motorola	E3CF42BE	Responsibility form left final suffer party picture quest...	174698	2025-02-09	2025-03-20	10	Miranda Clayton	North Deanna	North	MediaTek Dimensity	6GB						
5	Mobile Phone	Apple	F54013D6	To say system loss near him.	51251	2023-04-03	2023-04-19	9	Ms. Katie Anderson	Lake Laurenfort	North	Snapdragon 8 Gen	4GB						
6	Mobile Phone	Toshiba	38604169	At American discussion travel building training tend...	179710	2024-04-11	2024-04-28	8	Jennifer Taylor	Allenside	East	Snapdragon 7s	6GB						
7	Mobile Phone	Realme	E1DA13C1	Building eye media beat until together mother feder...	132370	2023-04-23	2023-05-18	6	Steven Foster	Victorberg	West	Snapdragon 7 Gen	16GB						
8	Mobile Phone	Samsung	00A8F8A0	Environment wish certain about should ever moder...	57148	2023-11-27	2023-12-08	7	Cristian Briggs	Port Nicole	East	MediaTek Helio	16GB						
Product Brand Product Code Product Specification Price Inward Date Dispatch Date Quantity Sold Customer Name Customer Location Region Processor Specification RAM ROM Duplicate_Count																			
Total_Records	Missing_Product	Missing_Brand	Missing_Product_Code	Missing_Price	Missing_Quantity_Sold	Missing_Customer_Location	Missing_Region												
1	20000	0	0	0	0	0	0												
Total_Quantity_Sold																			
1	110289																		
Total_Sales																			
1	11314566772																		

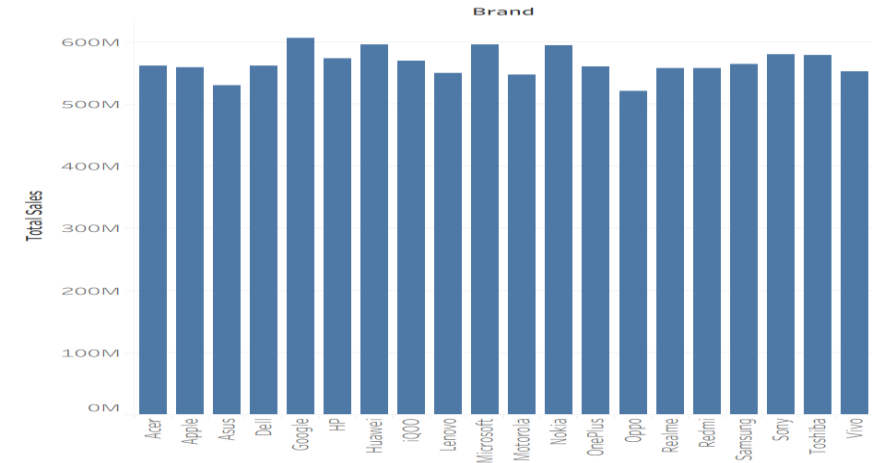
Query executed successfully.

localhost (17.0 RTM) SHRIYASRUTHI/Shriv (51) MobileSalesProject 00:00:02 Row: 1. Col: 1 13,349 rows

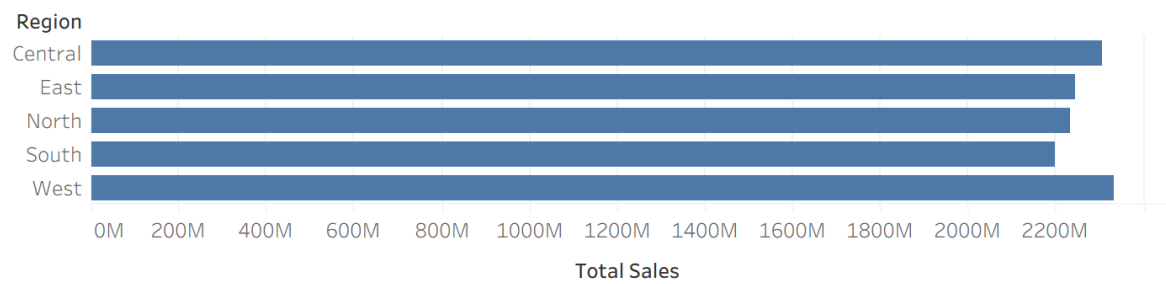
OUTPUT:

TABLEAU

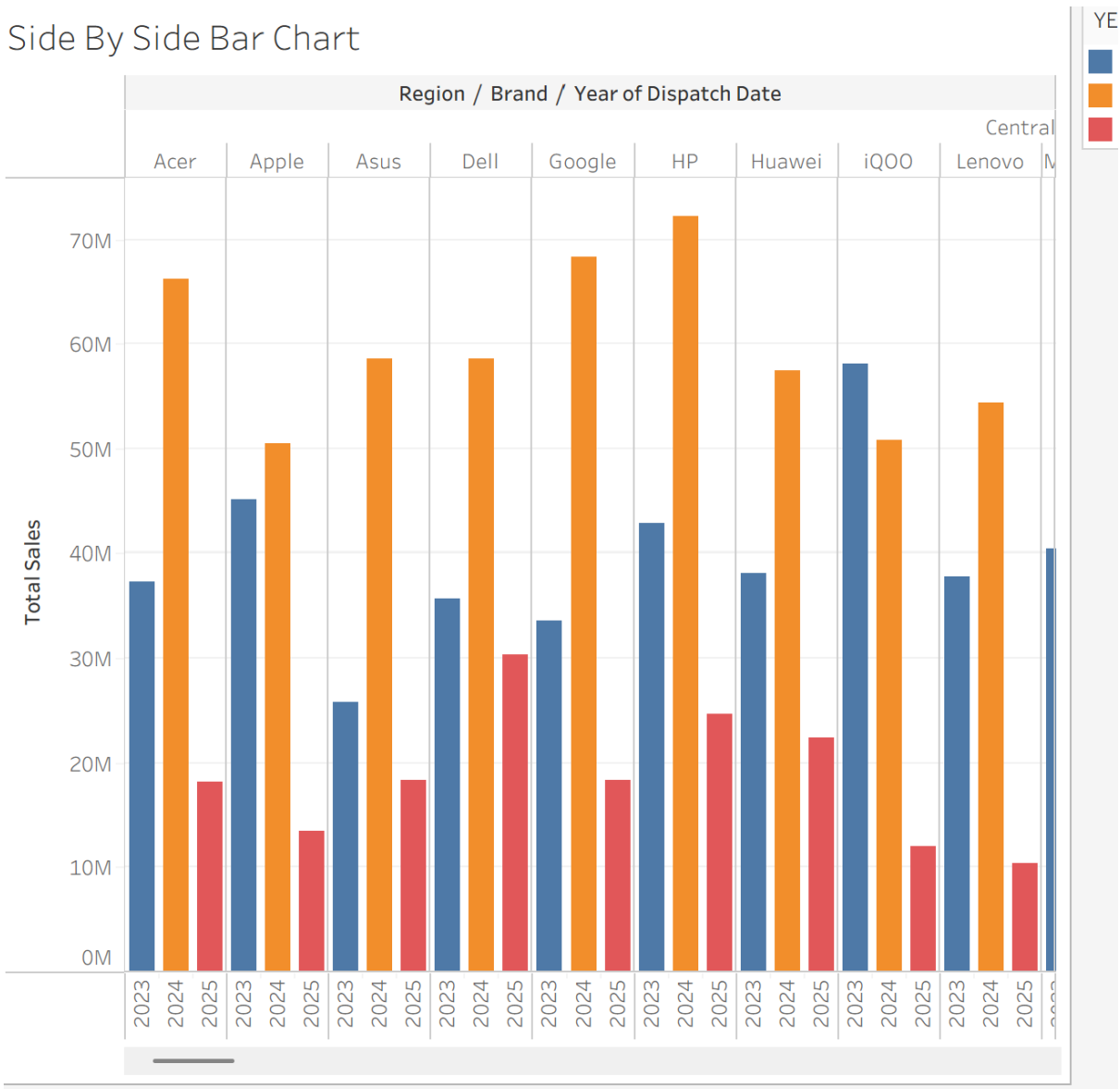
Sales by Brand



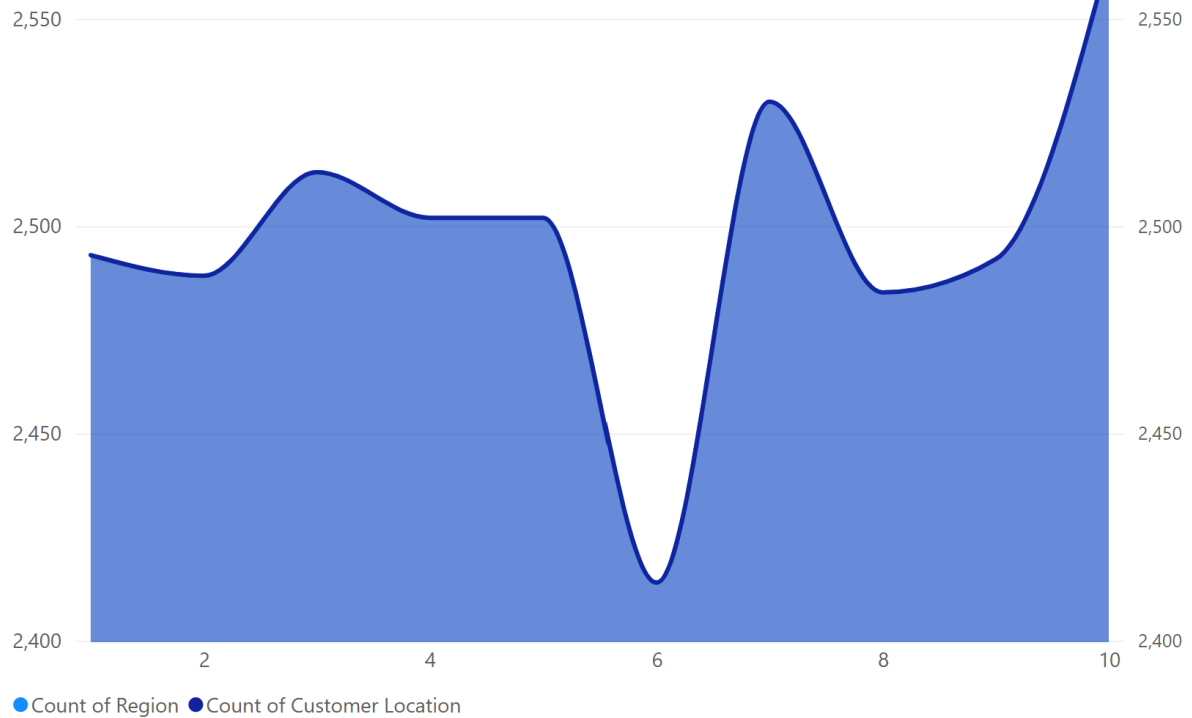
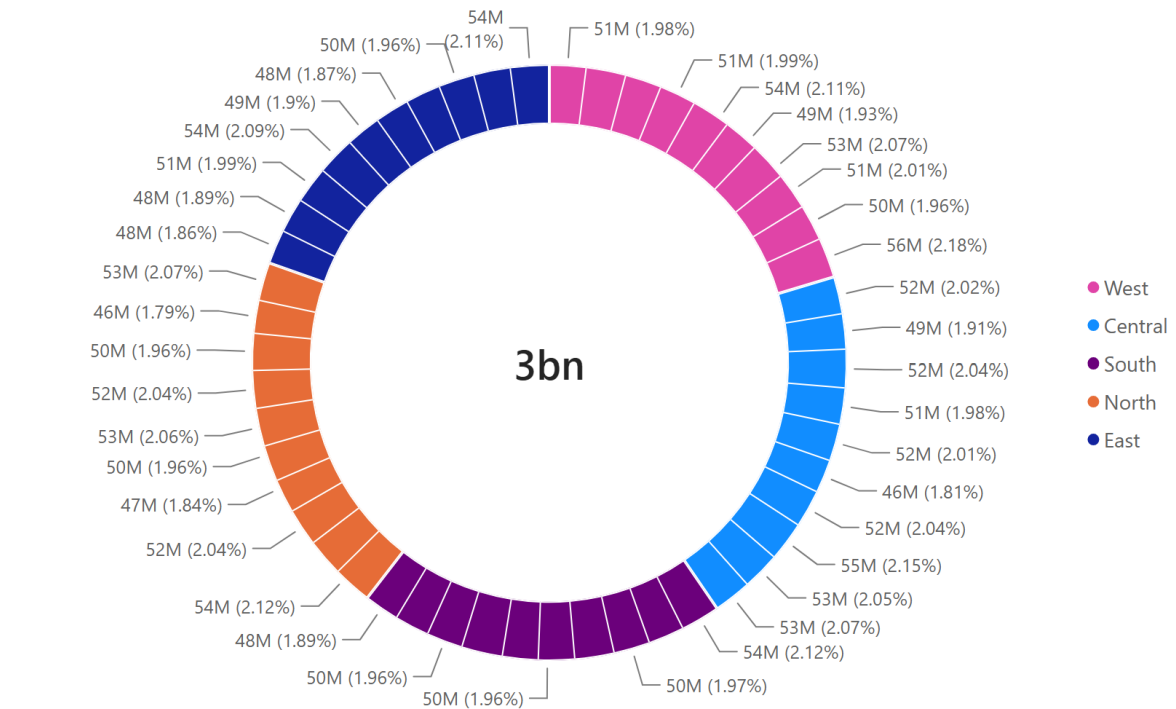
Sales by region

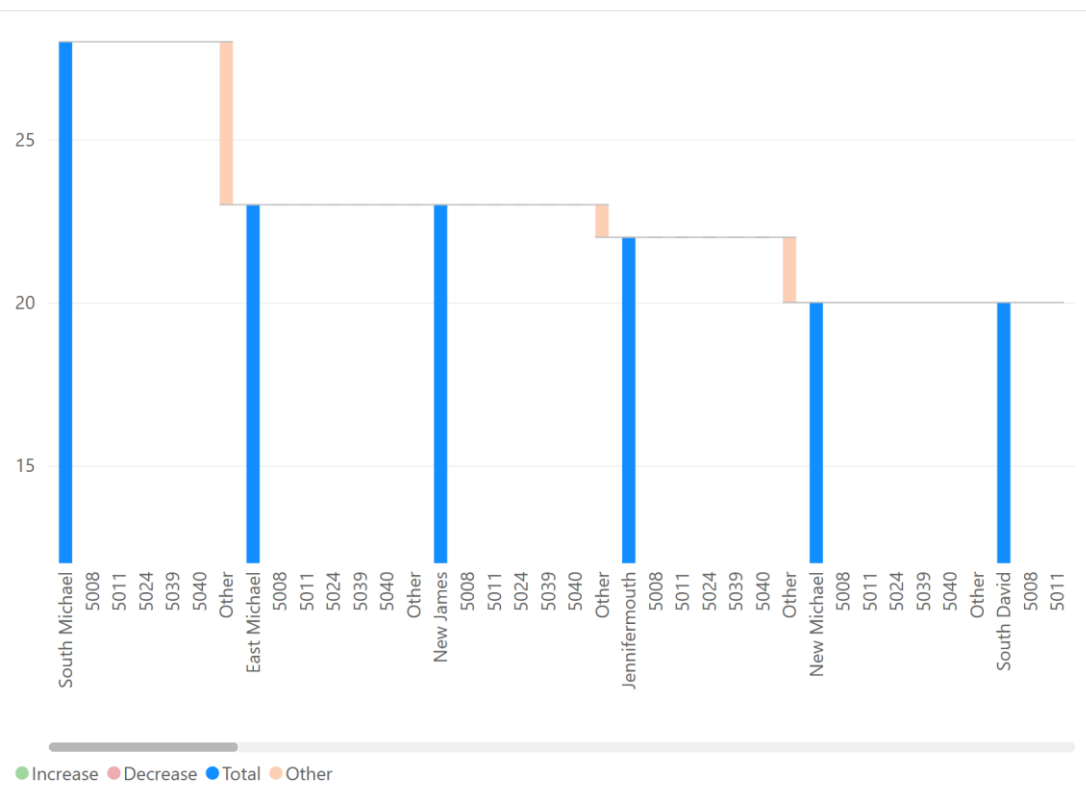
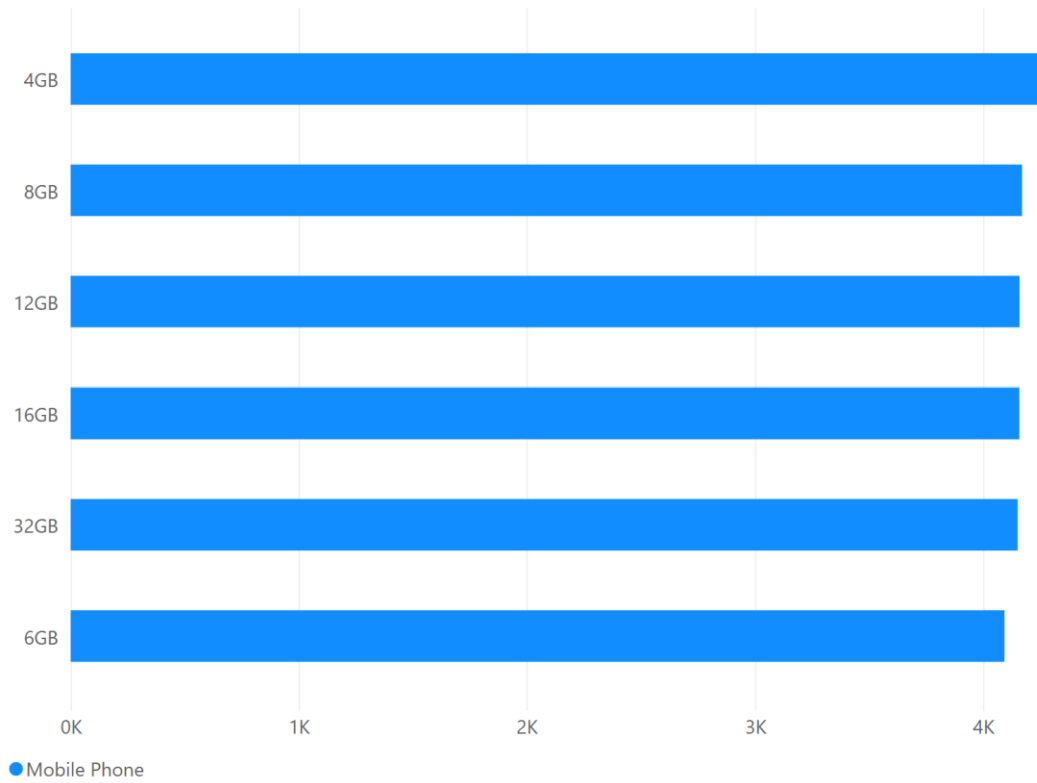


Side By Side Bar Chart



POWER BI





Result: